

Group 6

Gender Differences in Weight and Height

Research Question



- 1 Are male students heavier than female students?
- 2 Are male students taller than female students?
- 3 Is the difference statistically significant?

Methods



Independence

Welch's Two-Sample t-test

Normality

Homoscedasticity

Welch's t-test was validated and selected due to large sample sizes and independent random sampling

Summary



Male adolescents are significantly heavier and taller than females.

Data



Dataset : YRBS 2007

Variables

- Grouping1:Male(n=7510)
- Group 2: Female (n=6166)

Response Variables

- Weight (kg)
- Height (cm)

Sample Size

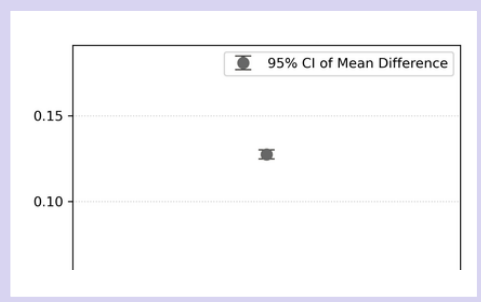
- Total : 14,041
- Weight available:
Male = 5510
Female = 6166
- Height available:
Male = 6490
Female = 6572

EDA Highlights

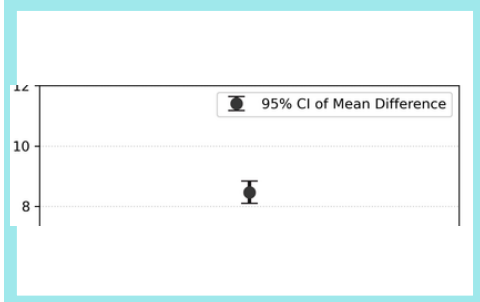


- ✓ Males are 12.75 cm taller than females on average.
- ✓ Males weigh 8.47 kg more than females on average.
- ✓ Males weigh visibly more than females overall.

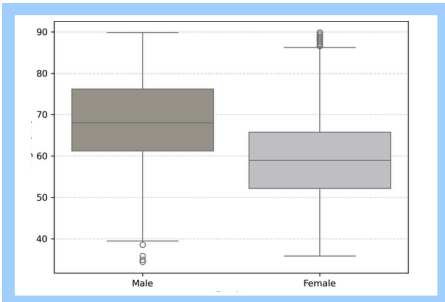
95% CI for Height Difference



95% CI for Weight Difference



Weight Distribution by Gender



Results



$$H_0 : \mu_{weight,M} = \mu_{weight,F}$$
$$H_1 : \mu_{weight,M} \neq \mu_{weight,F}$$

$$H_0 : \mu_{height,M} = \mu_{height,F}$$
$$H_1 : \mu_{height,M} \neq \mu_{height,F}$$



Weight (kg)

♂ $SD = 10.25$ ♀ $SD = 10.23$
♂ $\bar{X} = 60.23$ ♀ $\bar{X} = 68.70$
 $t = 44.62, p < 0.0001$ (Reject H_0)
Mean Difference = 8.47 kg
95%CI : [8.10, 8.84]kg
🔍 Males are significantly heavier.



Height (cm)

♂ $SD = 8$ ♀ $SD = 7$
♂ $\bar{X} = 176$ ♀ $\bar{X} = 163$
 $t = 92.31, p < 0.0001$ (Reject H_0)
Mean Difference = 12.75cm
95%CI : [12.47, 13.02]cm
🔍 Males are significantly taller.

ALL TESTS



- ✓ Reject H_0
- ✓ $p < 0.0001$
- ✓ Highly significant gender differences

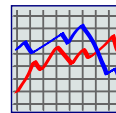
Effect Size

- Weight : d = 0.83
Large
- Height : d = 1.67
Very Large

Limitation



Observational data



Correlation only



Not causal